

Descriptions are all you need: Semantic Vectorization of Hierarchical Medical Knowledge (ICD-11) via Large Language Models

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Abstract

The 11th Revision of the *International Classification of Diseases* (ICD-11) serves as the global diagnostic standard established by the World Health Organization (WHO). However, **ICD-11** contains **limited and sparse descriptions** that challenge effective computational use. Current proprietary biomedical language models, such as Google’s Med-PaLM 2, demonstrate the capability to enhance medical textual representation but remain broadly inaccessible.

We leverage **Llama3-OpenBioLLM-70B** to systematically enrich ICD-11 descriptions, significantly improving their completeness and semantic density. We **generate** concise and rich **medical descriptions** structured around causes, symptoms, transmission, and diagnosis. The enhanced texts undergo rigorous linguistic, medical, and hierarchical validation, showing marked improvements in vocabulary diversity, readability, medical relevance, and structural consistency.

This work evaluates traditional **embedding methods** (TF-IDF, FastText) alongside advanced transformer-based models (PubMedBERT, BioBERT, GatorTron), highlighting their distinct strengths across biomedical tasks such as comorbidity detection, symptom-disease matching, and hierarchical accuracy. Contextual biomedical models demonstrate superior semantic retrieval capabilities, whilst traditional methods better capture direct lexical overlaps in clinical tasks.

Through this study, we provide the **first open-source vectorization of ICD-11**, which poses greater challenges than ICD-10 due to its **hierarchical, interconnected, and structurally complex architecture**. This resource enables large-scale biomedical retrieval and supports future clinical decision systems.

1 Introduction

The *International Classification of Diseases*, 11th Revision (ICD-11), developed by the World Health Organization (WHO), serves as the global diagnostic standard, organizing more than 40,000 entities and disease extensions within a complex and cross-referenced hierarchical structure. [29] Despite its breadth, ICD-11 descriptions suffer from significant information gaps, hindering their effective use in computational applications such as semantic search and automated coding. [Fig. 1]

Recent advances in large biomedical language models such as Google’s Med-PaLM 2 and AMIE have demonstrated the feasibility of automating medical knowledge representation. [21] [23] However, these models remain proprietary and inaccessible for further applications.

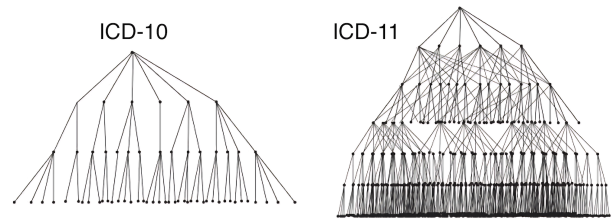


Figure 1: Foundation structure comparison between the old ICD-10 and the new ICD-11 medical taxonomy.

To address this limitation, we leverage Llama3-OpenBioLLM-70B, an open-source biomedical language model known for its robust performance, to systematically enrich ICD-11 descriptions. [24] In this paper, we first explore existing research in embedding ICD codes and automated medical classification (Section 2). We then detail our method for data extraction, description generation, and validation procedures (Section 3), to emphasize the linguistic, medical, and hierarchical aspects of the newly generated texts. Subsequently, we outline our experimental setup for evaluating various embedding methods, including traditional (TF-IDF, FastText) and contextual transformer models (PubMedBERT, BioBERT, and GatorTron).

Our evaluation benchmarks, including comorbidity scores, symptom-disease matching, and hierarchical consistency, are discussed thoroughly (Section 4). Finally, we summarize our research’s contributions, highlight its limitations, and suggest directions for future work (Section 5).

2 Related Works

Previous efforts in automating ICD coding primarily focus on improving disease representation and retrieval. Feng et al. (2024) demonstrate that straightforward methods like TF-IDF and cosine similarity still effectively match medical narratives to ICD-11 codes, but their approach lacks contextual awareness, limiting semantic precision [5].

Lee et al.’s ICD2Vec (2023) introduces vector embeddings for ICD codes to capture disease similarity and comorbidity patterns. Their work provides the foundation for our work, as it is tailored for ICD-10 and does not handle natural language queries [10]. Chen and Xu (2020) integrate WordNet to enrich ICD-11 descriptions, enhancing semantic alignment without addressing the original ICD texts’ inherent sparsity. [4] Conversely, Klotzman (2024) finds embedding-based methods superior to large language models (LLMs) like GPT-4 in automatic ICD coding. [8]

Recent transformer-based approaches further highlight the potential of contextual embeddings. PLM-ICD (Wang et al., 2023) successfully applies pretrained transformer models for clinical text coding, though focused on ICD-9 rather than ICD-11, and primarily on long-form documents rather than interactive queries. [27] DiLBERT provides specialized embeddings for disease-related tasks, outperforming general-domain embeddings in semantic medical tasks. [31]

Finally, the ICDBigBird model combines Graph Convolutional Networks (GCN) and transformers to manage lengthy clinical texts, achieving robust ICD assignment performance but at the cost of significant computational overhead, unsuitable for real-time interactions. [12]

Overall, existing research underscores the importance of specialized biomedical embeddings but does not comprehensively address the limitations of ICD-11’s description sparsity or provide optimized solutions for real-time natural language querying.

3 Experiment

3.1 Data Extraction

We load ICD-11’s data using WHO’s API, retrieving entities with Uniform Resource Identifiers (URIs). [28] Given the database’s tree structure, we employ a breadth-first search algorithm to crawl each chapter, storing entries as JSON files. Then, we normalize these fields into a table with 19 columns and 13,960 rows, comprising 10,678 diseases (76%) and 3,282 intermediary classification entities (24%). Description length averages at 154 characters, with a median of 0. Moreover, 7,066 entries have empty descriptions, and our custom completeness score, evaluating the presence of terms for causes, symptoms, transmission, and diagnosis, averages 14%. [B.1]

3.2 Data Generation

To overcome description deficiency, we generate new data using Llama3-OpenBioLLM-70B (LOB), a state-of-the-art biomedical language model that outperforms proprietary alternatives like MedPalm2 and GPT-4 on biomedical benchmarks. [24][14][22] We instruct LOB to generate medical descriptions following a standardized order (disease causes, symptoms, transmission, and

diagnosis), at a temperature of 0.2 for deterministic reproducibility, with output capped at 800 tokens to favour information density. [B.2] The generated descriptions average 800 characters, with none being empty, and a mean completeness score of 78%, demonstrating improved utility for vectorization.

3.3 Data Analysis

Linguistic quality. Generated descriptions have superior vocabulary richness with lower cross-chapter variance in Type-Token Ratio (TTR) and Lexical Diversity (LD) scores. [16] Furthermore, their Flesch-Kincaid score corresponds to academic-level readability (range 12 to 15.5), contrasting with the original’s intermediate level at an average of 8.9. [9] Finally, low BLEU scores (ranging 0.01-0.055) and moderate ROUGE-1 and ROUGE-L scores (0.11-0.29) point to substantial information novelty. [15][11][B.3]

Medical validation. Using a dictionary of 22 medical operators, we find that generated descriptions are 6 times more causal across chapters 1-19, while preserving variance patterns from the original descriptions. Moreover, the composition of these descriptions follows the structured prompt and aligns with medical intuition. For instance, infectious diseases descriptions (Ch. 1) dedicate more text (25% vs. 5% average) to transmission, while external cause diseases (Ch. 22) emphasize causes (62% vs 41% average). Interestingly, LOB exhibits an average 2% drift towards generating unprompted treatment information. [1] We confirm the medical relevance by finding that descriptions cluster along five semantic axes, consistent with established medical theory. [B.4]

Hierarchical analysis. ICD-11 has a leaf-dense structure, with an average node depth of 3.54 and a branching factor of 4.25. Using all-MiniLM-L6-v2 embeddings, we confirm that content is preserved across hierarchical levels (mean cosine similarity 0.636), and semantic drift occurs in diverse subcategory nodes as expected. [26] Head/Face conditions dominate with 5,438 diseases and 724,800 description words, followed by Brain (1,494 diseases, 210,702 words) reflecting how medical classification prioritizes complex body areas related to consciousness, cognition, and social interaction. [B.5]

3.4 Embedding Models

We generate text embeddings using seven different models, partially following the approach of Lee et al. (2023) [10]. Two are traditional methods, TF-IDF and FastText, and the remaining five are pre-trained transformer encoders: the general-domain BERT (bert-base-uncased) and four biomedical variants—BioBERT (PubMed abstracts), BioClinicalBERT (MIMIC-III clinical notes), PubMedBERT (from-scratch training on PubMed abstracts and full texts [6]), and GatorTron (82 billion-token clinical notes corpus [30]).

We extend beyond the original ICD2Vec paper by including PubMedBERT, as it uniquely combines domain-specificity with being trained from scratch on biomedical text (unlike BioBERT, which was initialized from BERT). PubMedBERT’s architectural choice prevents potential domain conflicts and shows superior performance in biomedical tasks. BERT serves as a general baseline, and its biomedical variants offer domain specialization through their training data.

3.5 Experimental Setup

Our embedding generation pipeline consists of several steps. First, we append structured ICD-11 metadata to the generated descriptions, including diagnostic synonyms, related conditions, and hierarchical relationships. Then, we apply text preprocessing using the NLTK package, including lowercase conversion, special character removal, and whitespace normalization. Next, we set up our models. We specify TF-IDF with a vocabulary of 3000 features and apply SVD reduction to match BERT’s hidden layer size. We configure FastText with a skip-gram model and a window size of five. For transformer models, we leverage their pre-trained weights. All models process texts in batches, with proper attention masking for transformers. We generate embeddings using mean pooling over the last hidden states, maintaining consistent 768-dimensional vectors across all approaches for fair comparison.

4 Results & Discussion

4.1 Intrinsic Evaluation

We assess the quality of the resulting embeddings across the seven models in the original 768-dimensional space by grouping by ICD-11 chapters and computing silhouette scores, Calinski-Harabasz indices, and Davies-Bouldin scores. [19][2][17] The best resulting embeddings are those produced by GatorTron (silhouette score 0.0315), PubMedBert (0.0215), and FastText (0.0134). [C.1] Additionally, we evaluate the quality of embeddings in a 2-dimensional space using t-SNE and UMAP visualisations, grouping by chapter. [Fig. 2][25][13]

We also compute trustworthiness and continuity measures, which complementarily measure whether neighbourhood relationships are preserved during dimensionality reduction. GatorTron and FastText again surpass other models, while BERT improves and ties with PubMedBert. [C.2]

4.2 Comorbidity Score Benchmark

A comorbidity score quantifies the mortality risk associated with having multiple diseases simultaneously. Lee et al. (2023) [10] use comorbidity scores of disease pairs to evaluate ICD-10 embeddings. Based on prior studies, they argue that more similar diseases tend to have higher

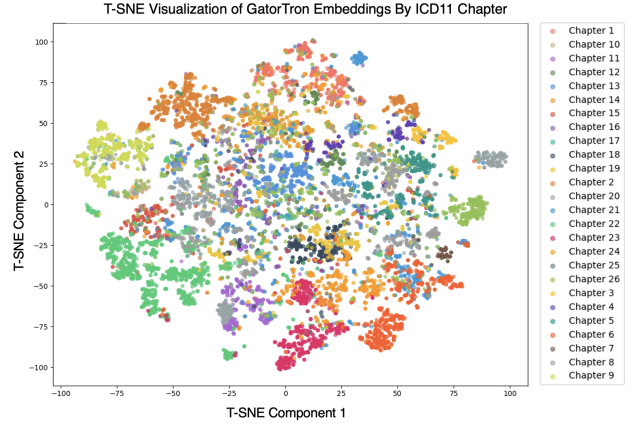


Figure 2: t-SNE low-dimensional visualization of the embeddings generated by GatorTron, grouped by ICD-11 chapter.

comorbidity, likely because they affect the same organ system and worsen each other. Lee et al. proposed that comorbidity scores should correlate positively with the cosine similarity of disease embeddings. This correlation coefficient, which we denote as the comorbidity benchmark score, acts as a metric for embedding quality. Supporting this idea, Schneider et al. [20] found that the correlation between comorbidity scores and semantic similarity of diseases reaches 0.75, indicating a strong positive relationship and validating the metric’s use.

As the UK Biobank data used by Lee et al. is not publicly accessible, we use comorbidity scores for 803 disease pairs from the study by Roque et al. [18], which is based on data from over 5,000 Danish patients.

Our results on the comorbidity score align with Lee et al.: TF-IDF performs best (0.28), followed by GatorTron (0.19) and PubMedBERT (0.17). [10][C.3] TF-IDF likely performs the best because ICD-11 codes of diseases with high comorbidity often share exact terms, which the model captures directly. In turn, contextual models may cluster diseases by topic, reducing alignment. Notably, the comorbidity benchmark scores of all models are weak, which suggests limitations in the embedding quality of all models. [Tab. 1]

Model	Comorbidity Benchmark Score
TF-IDF	0.2801
FastText	0.0966
BERT	0.1332
BioBERT	0.1260
BioClinicalBERT	0.0972
PubMedBERT	0.1706
GatorTron	0.1886

Table 1: Comorbidity benchmark score across models

4.3 Symptom-Disease Matching

Chapter 21 of the ICD-11 classification covers disease symptoms (e.g., cough, asphyxia).

We use model embeddings for each of the 940 symptoms to identify the disease with the highest cosine similarity, interpreted as the disease most likely associated with that symptom. Lee et al. use a similar method and manually judge whether the resulting symptom–disease pairs are medically reasonable. [10] We confirm the validity of the results with visual inspection and compare models through a numerical analysis. [C.4] TF-IDF shows a much lower average cosine similarity (0.62) than all other models (each above 0.96), indicating that its embeddings are sparser and less uniformly scaled. We also measure how often two models predict the same top disease for a symptom. On average, TF-IDF’s predictions match those of other models in only 19.4% of symptoms, indicating its less aligned behaviour. In contrast, contextual and medical-trained models like PubMedBERT, BioBERT, and GatorTron produce more consistent results, agreeing on the top disease for about 35% of symptoms.

4.4 Encyclopedia Definition Benchmark

This evaluation assesses whether language models accurately match previously unseen disease definitions to corresponding ICD-11 codes. We extract 718 single-word disease terms from ICD-11, retrieve definitions using Merriam-Webster’s Medical Dictionary API, and filter to 378 unambiguous ICD matches. [3]

ICD-11 codes follow a hierarchical structure, where codes sharing initial symbols indicate related diseases. We embed each definition, identify the top 3 most similar ICD-11 codes via cosine similarity, and measure shared initial symbols (1–4) to evaluate hierarchical accuracy.

Results show that contextual BERT-based models outperform traditional methods in initial symbol matches (82–92% vs. TF-IDF 31.48%, FastText 16.67%). Crucially, accuracy decay varies widely: PubMedBERT retains accuracy well from 92.33% (1 symbol) to 75.13% (4 symbols), whereas GatorTron sharply drops from 50.79% to 2.91%. [C.5] This happens likely because GatorTron was trained mainly on clinical notes, with less exposure to structured ICD-like text than PubMedBERT or BioBERT, making it weaker at learning code patterns and hierarchical links. Traditional embeddings show even steeper declines—TF-IDF from 31.48% to 1.06%, FastText from 16.67% to 0.79%. Thus, capturing ICD-11’s hierarchical structure demands sophisticated contextual embeddings pre-trained on medical texts.

5 Conclusions

This work demonstrates that the 11th version of the International Classification of Diseases can (i) be enriched using Llama3-OpenBioLLM-70B, improving completeness from 14% to 78%, and (ii) be effectively vectorized using embedding pipelines and state-of-the-art biomedical encoders. We are the first open-source research

community to vectorize the nuanced ICD-11 medical taxonomy. Our contribution is twofold.

First, we provide an LLM-augmented ICD-11 dataset that has been rigorously validated across multiple axes: linguistically, through type-token ratio, lexical diversity, and Flesch-Kincaid readability; semantically, via BLEU and ROUGE metrics; and medically, by measuring causal marker count, disease-component composition, and alignment with clinical axes.

Second, using seven different models, we develop a vector index of the augmented ICD-11, which has been carefully evaluated across diverse biomedical tasks. The results highlight the complementary strengths of traditional and contextual models. On the one hand, contextual biomedical models, particularly PubMedBERT, significantly outperform conventional methods in tasks requiring deep semantic understanding. These include symptom-disease matching, encyclopedia definition retrieval (improving recall from 31.48% with TF-IDF to 92.33% with PubMedBERT for 1-symbol prefix match), and preserving hierarchical structure. Contrarily, TF-IDF demonstrated superior performance for comorbidity detection ($\rho=0.28$), and FastText’s embeddings rank among the highest in quality, both in low- and high-dimensional spaces. Our findings align with those from other works, even under different datasets, as with our comorbidity benchmark and results from [10]. These enriched vector representations pave the way for advanced semantic retrieval systems and clinical decision support tools based on ICD-11.

In addition to the evaluation methods discussed in Section 4, we also compare our models on AI-generated prompts written in the style of everyday language, rather than professional medical terminology [C.6]. The models perform poorly on this test. A likely explanation is that contextual models like GatorTron and PubMedBERT are primarily trained on texts written by medical professionals and therefore struggle to interpret informal or non-technical language used by ordinary people when describing medical concerns.

This limitation points to three directions for future work: (i) integrate graph neural networks (GNNs) trained with self-supervised objectives (e.g., link prediction) to capture hierarchical and cross-disease relationships better; (ii) utilize cutting-edge foundation models such as GPT-4.1, Claude Sonnet 4, or Grok 3 to enhance embedding quality; and (iii) explore scaling laws by varying embedding dimensions, model capacity, and input length to optimization effectiveness. [7]

Together, these improvements aim to make ICD-11 embeddings more representative of biological mechanism and disease interactions and adaptable to questions posed in everyday language, ultimately supporting many real-world applications across multiple layers of society.

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Appendix

A General ICD-11 Information

A.1 Chapter list

1. Certain infectious or parasitic diseases
2. Neoplasms
3. Diseases of the blood or blood-forming organs

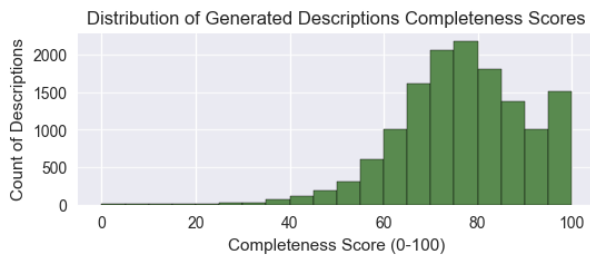
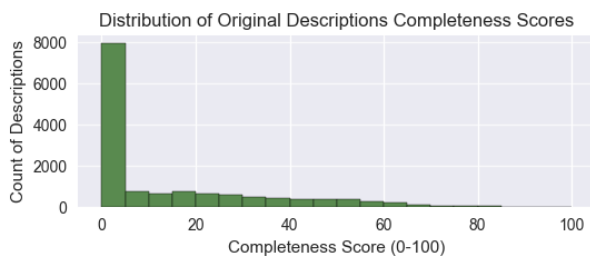
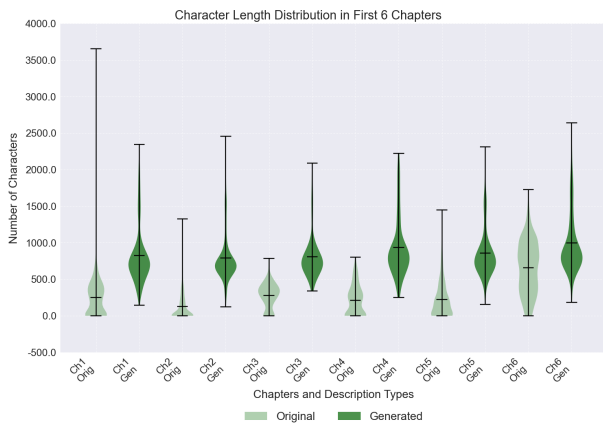
4. Diseases of the immune system
5. Endocrine, nutritional or metabolic diseases
6. Mental, behavioural or neuro-developmental disorders
7. Sleep-wake disorders
8. Diseases of the nervous system
9. Diseases of the visual system
10. Diseases of the ear or mastoid process
11. Diseases of the circulatory system
12. Diseases of the respiratory system
13. Diseases of the digestive system
14. Diseases of the skin
15. Diseases of the musculoskeletal system or connective tissue
16. Diseases of the genitourinary system
17. Conditions related to sexual health
18. Pregnancy, childbirth or the puerperium
19. Certain conditions originating in the perinatal period
20. Developmental anomalies
21. Symptoms, signs or clinical findings, not elsewhere classified
22. Injury, poisoning or certain other consequences of external causes
23. External causes of morbidity or mortality
24. Factors influencing health status or contact with health services
25. Codes for special purposes
26. Supplementary Chapter Traditional Medicine Conditions

B Data Processing and Analysis

B.1 Character Length and Completeness Score

We devise a completeness score metric to quantify the information richness of each description. The score is built by counting the presence of terms relating to essential fields of a disease, including causes, symptoms, transmission, and diagnosis, and then normalising to 0 and 1, with higher values indicating a more comprehensive description. Computing the score for each description

and plotting the distribution of scores, we confirm our hypothesis: with a mean completeness score of 14 out of 100, ICD-11 descriptions alone are not sufficient for developing rich embedding representations of all diseases. This poses a significant challenge to our goal of vectorising information in ICD-11.



```
aspects = {
  'overview': [
    'is a', 'are a', 'defined as', 'refers to', '
    condition', 'disease', 'disorder',
    'syndrome', 'illness', 'characterised', '
    affecting', 'occurs when', 'type of',
    'form of', 'group of', 'category of', 'class
    of', 'classification', 'entity',
    'pathology', 'abnormality', 'problem', '
    health issue', 'medical', 'clinical',
    'pathological', 'involves', 'designates', '
    term for', 'describes', 'relating to',
    'concerning', 'pertaining to', 'associated
    with', 'manifested by', 'feature of'
  ],
  'causes': [
    'cause', 'aetiology', 'risk factor', 'due to',
    'result of', 'caused by',
    'trigger', 'associated with', 'genetic', '
    infection', 'bacteria', 'virus',
    'fungus', 'parasite', 'hereditary', '
    environmental', 'exposure', 'origin',
    'source', 'underlying', 'predisposing', '
    development', 'derived', 'acquired',
    'congenital', 'induced by', 'secondary to', '
    following', 'after', 'linked to',
    'consequence of', 'sequela', 'complication',
    'outcome', 'related to', 'response',
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    'autoimmune', 'toxin', 'chemical', 'radiation',
    'trauma', 'injury', 'defect',
    'mutation', 'deficiency', 'excess', 'pathogen',
    'agent', 'mechanism'
  ],
  'symptoms': [
    'symptom', 'sign', 'present with', 'manifests
    as', 'characterised
    presentation', 'clinical feature', 'pain', '
    fever', 'swelling', 'fatigue',
    'discomfort', 'ache', 'loss', 'impair', '
    dysfunction', 'manifestation',
    'complaint', 'report', 'sensation', '
    experience', 'feeling', 'suffer',
    'exhibit', 'display', 'show', 'demonstrate',
    'express', 'appear', 'develop',
    'progress', 'worsen', 'improve', 'resolve', '
    persist', 'severe', 'mild',
    'moderate', 'acute', 'chronic', 'subjective',
    'objective', 'visible',
    'palpable', 'tender', 'redness', 'rash', '
    lesion', 'weakness', 'numbness',
    'tingling', 'stiff', 'rigid', 'sore', '
    uncomfortable', 'irritated', 'inflamed'
  ],
  'transmission': [
    'transmit', 'spread', 'contagious', '
    infectious', 'vector', 'airborne',
    'waterborne', 'bloodborne', 'contact', '
    carrier', 'contamination',
    'infected', 'transmitted by', 'spreads
    through', 'communicable',
    'person-to-person', 'direct', 'indirect', '
    faecal-oral', 'droplet',
    'aerosol', 'bodily fluid', 'saliva', 'blood',
    'sexual', 'vertical',
    'maternal', 'perinatal', 'breast milk', '
    transplacental', 'nosocomial',
    'hospital-acquired', 'community-acquired', '
    endemic', 'epidemic', 'pandemic',
    'zoonotic', 'animal', 'insect', 'mosquito', '
    tick', 'flea', 'lice', 'parasite',
    'reservoir', 'host', 'incubation', 'shedding'
  ],
  'diagnosis': [
    'diagnos', 'test', 'examination', 'detection',
    'screening', 'identified by',
    'laboratory', 'imaging', 'scan', 'blood test',
    'confirmed by', 'biopsy',
    'differential', 'assess', 'evaluation', '
    workup', 'investigation', 'findings',
    'clinical', 'physical', 'history', 'criteria',
    'marker', 'indicator', 'biomarker',
    'analyse', 'monitor', 'observe', 'determine',
    'identify', 'confirm',
    'rule out', 'exclude', 'distinguish', '
    characterise', 'categorise', 'radiograph', 'x-
    ray', 'CT', 'MRI', 'ultrasound', 'endoscopy', '
    culture', 'PCR',
    'serology', 'antibody', 'antigen', 'genetic',
    'molecular', 'biopsy', 'specimen',
    'sample', 'tissue', 'fluid', 'urine', 'stool',
    'cerebrospinal', 'CSF'
  ],
  'treatment': [
    'treatment', 'therapy', 'intervention', '
    management', 'medication', 'drug',
    'surgery', 'therapeutic', 'treated with', '
    managed by', 'cure', 'alleviate',
    'relieve', 'regimen', 'protocol', 'approach',
    'prevented', 'resolution',
    'heal', 'treat', 'manage', 'control', 'care',
    'support', 'palliative', 'remedial',
    'procedure', 'operation', 'technique', '

```



```

method', 'antibiotic', 'antiviral',
58     'antifungal', 'antimicrobial', 'steroid', '
hormone', 'vaccine', 'immunisation', 'prevention',
    'prophylaxis', 'recovery', 'rehabilitation', '
physical therapy',
59     'occupational therapy', 'cognitive therapy',
'behavioural therapy', 'psychotherapy',
60     'counselling', 'dietary', 'nutrition', '
lifestyle', 'modification', 'complementary',
61     'alternative', 'traditional', 'response', '
prognosis', 'outcome', 'remission',
62     'follow-up', 'monitoring', 'maintenance', '
long-term'
63 ],
64 # Additional aspect to capture complication and
prognosis information
65 'prognosis': [
66     'prognosis', 'outlook', 'course', '
progression', 'complication', 'sequela',
67     'mortality', 'morbidity', 'survival', 'fatal',
'life expectancy', 'chronic',
68     'relapse', 'recur', 'remission', 'resolve', '
improve', 'deteriorate', 'worsen',
69     'permanent', 'temporary', 'reversible', '
irreversible', 'disability', 'impairment',
70     'function', 'quality of life', 'burden', '
risk', 'chance', 'likelihood', 'percent',
71     'rate', 'common', 'rare', 'frequent', '
occasional', 'variable', 'unpredictable'
72 ]
73 }

```

Listing 1: Python Code Definition of Dictionary used for computing Completeness Score

B.2 Llama3-OpenBioLLM Python Setup

```

1 from openai import OpenAI
2 from dotenv import load_dotenv
3 import os
4
5 load_dotenv()
6
7 SYSTEM_MESSAGE = """
8 Your name is Llama3-OpenBioLLM-70B. You are an expert
   and experienced from the healthcare and
   biomedical domain with extensive medical
   knowledge.
9 Your mission is to provide comprehensive, technical,
   and accurate medical description descriptions of
   diseases and disease categories.
10 The user will input the name of a disease or the name
   of a category of diseases.
11 You will provide the description of the query.
12 Always structure the sentences of your response in
   this order: overview, causes, symptoms,
   transmission, diagnosis.
13 Write full sentences, using a concise and clear
   language.
14 """
15
16 def get_medical_definition(query, temperature=0.2):
17     """
18     Get a medical definition using the Nebius API and
19     Llama3-OpenBioLLM-70B model.
20     """
21     api_key = os.getenv("NEBIUS_API_KEY")
22     if not api_key:
23         raise ValueError("NEBIUS_API_KEY not found in
24         environment variables")
25
26     client_nebius = OpenAI(
27         base_url="https://api.studio.nebius.com/v1/",

```

```

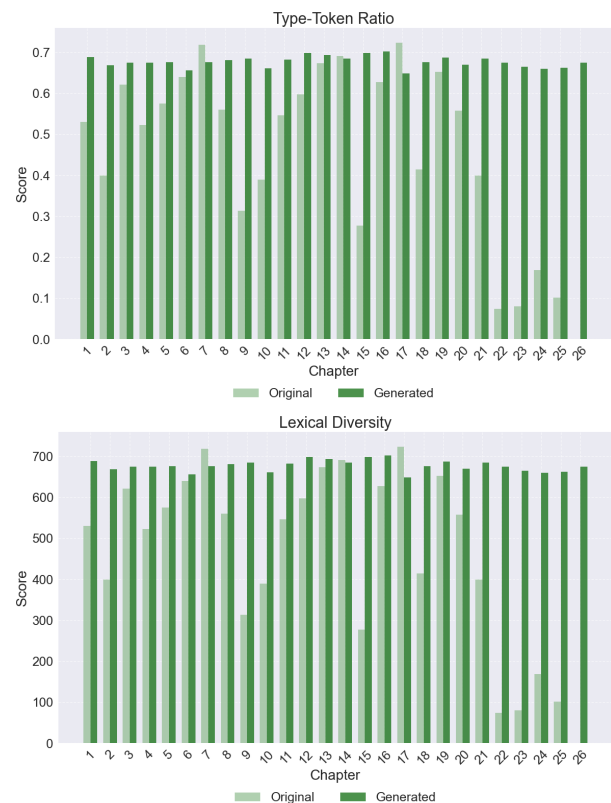
26     api_key=api_key)
27
28     try:
29         response = client_nebius.chat.completions.
30         create(
31             model="aaditya/Llama3-OpenBioLLM-70B",
32             temperature=temperature,
33             messages=[
34                 {"role": "system", "content":
35                 SYSTEM_MESSAGE},
36                 {"role": "user", "content": f"
37                 Describe {query}?"}
38             ],
39             max_completion_tokens=800
40         )
41
42         return response.choices[0].message.content
43
44     except Exception as e:
45         print(f"Error getting medical definition: {
46         str(e)}")
47         raise

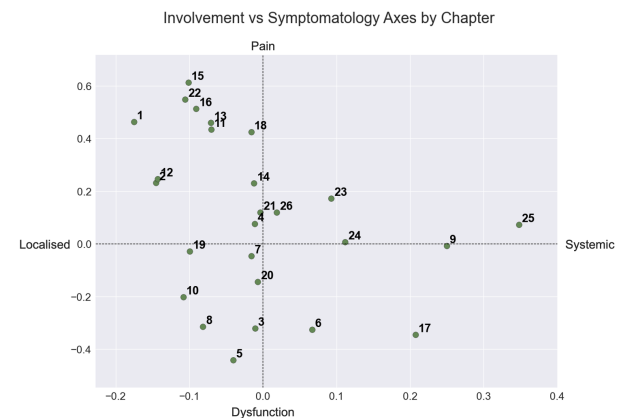
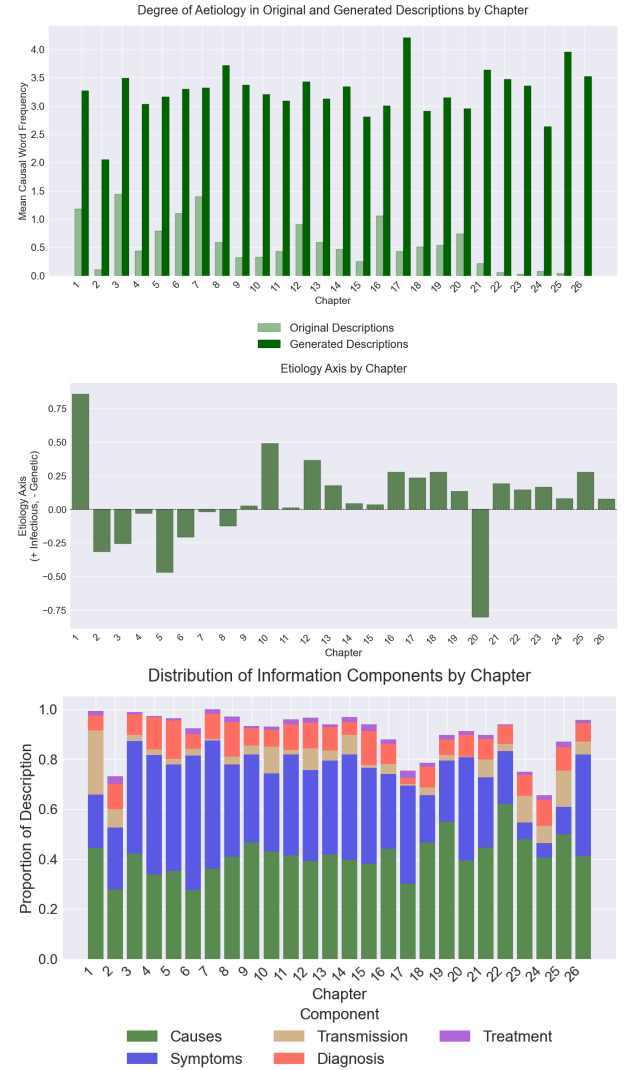
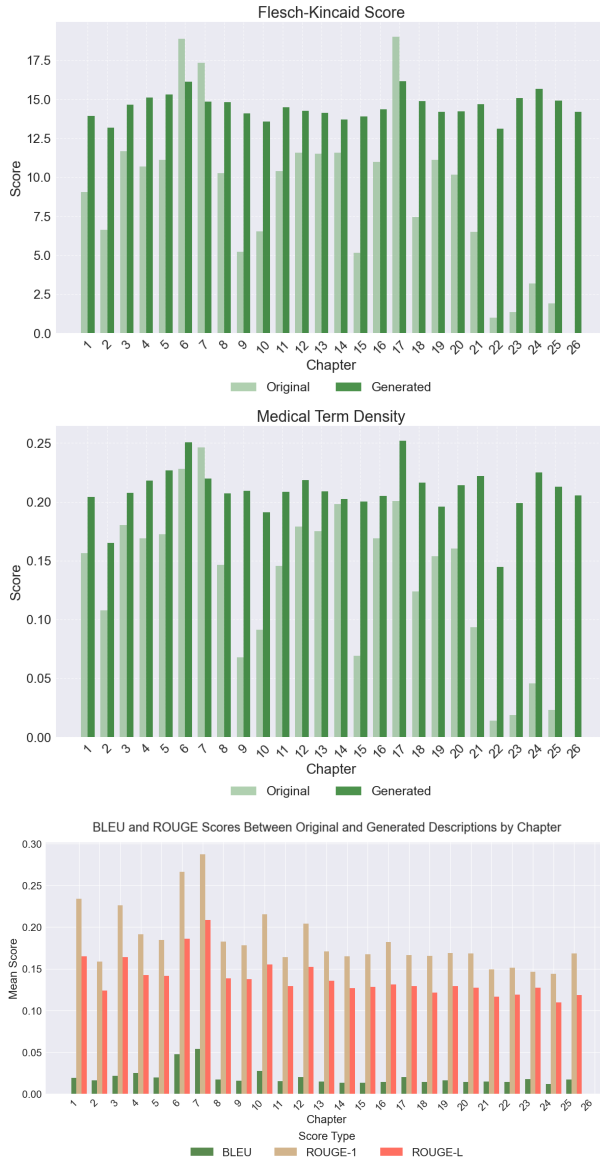
```

Listing 2: Python Code for Medical Definition Generation

B.3 Linguistic quality

Information richness relates to the text's linguistic quality. As such, we only apply light cleaning with lowercase and punctuation removal to preserve semantic meaning and expression. Then, we compute various metrics and compare original and generated descriptions across ICD-11 chapters.





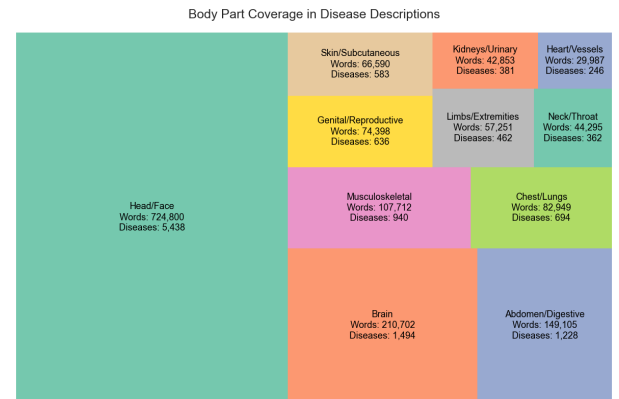
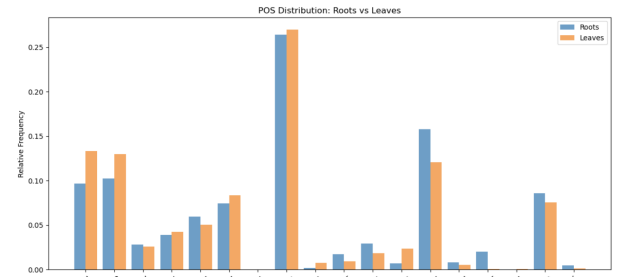
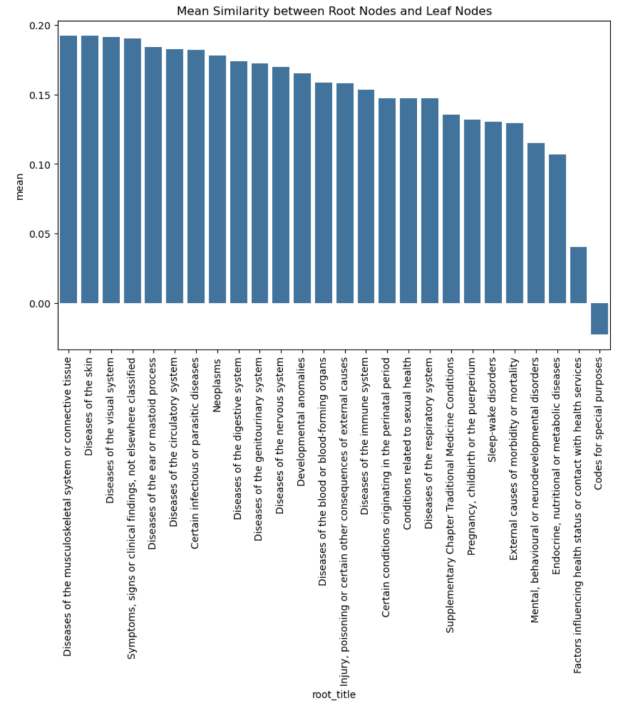
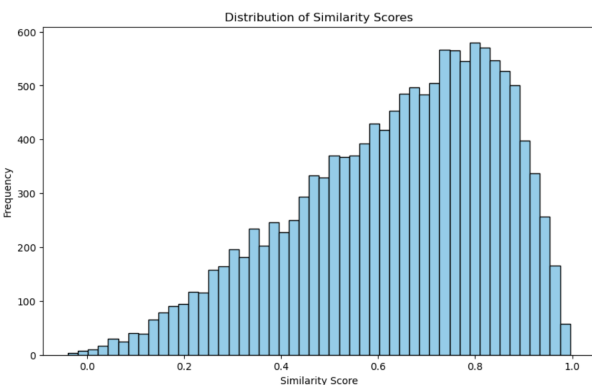
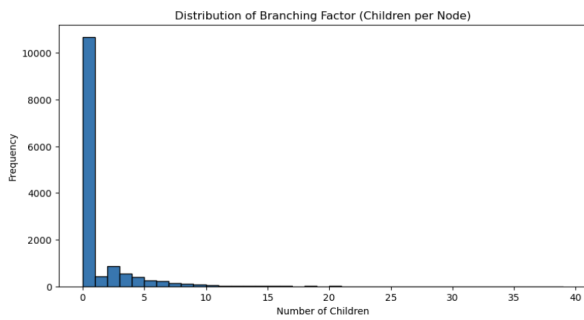
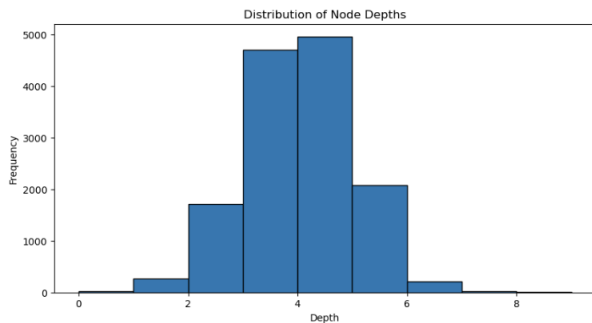
B.4 Medical validation

We run a series of dictionary-based tests specifically designed to validate the content of the generated descriptions from a medical perspective. Dictionary-based methods rely on predefined sets of keywords and phrases that are related to specific medical concepts. Their advantages are explicit control over the evaluation criteria, direct interpretation, and test speed on the text corpus. The main disadvantage is that the results depend heavily on the contents of the dictionaries, which demands additional effort to ensure that the dictionary is capable of classifying any expression.

B.5 Hierarchical analysis

Given the structure of the dataset—composed of individual nodes representing ICD-11 codes situated at various levels within the original hierarchy—we undertook a detailed analysis of the graph-like structure implied by the data and how semantic and linguistic information propagates from higher-level categories (root nodes) to more granular clinical entries (leaf nodes). We look at node depth, the number of edges from the root node to that point, branching factor, the number of children each node has, parent-child description similarity, root-leaf description similarity across chapters, and root-leaf POS distribution.

Finally, we produce a nested treemap to analyze anatomical distribution of disease descriptions across the human body. We again leverage comprehensive dictionaries of anatomical terms, mapping specific keywords to eleven major body systems (e.g., "cerebral," "neural" for Brain; "cardiac," "vascular" for Heart/Vessels). We also include the total word count dedicated to describing conditions affecting each body part, and the number of distinct diseases associated with each anatomical region.



C Embeddings Evaluation

C.1 Intrinsic Evaluation Results

See Table 2, Table 3, and Table 4.

Model	Silhouette Score	Calinski-Harabasz Score	Davies-Bouldin Score
BERT	0.0859	171.5505	4.0329
BioBERT	0.0164	119.8868	4.1165
BioClinicalBERT	0.0839	123.0997	4.2670
PubMedBERT	0.0215	135.2616	3.8855
TF-IDF	0.0013	45.8275	5.6999
FastText	0.0134	230.7401	3.4326
GatorTron	0.0315	135.8351	3.8102

Table 2: Embedding Quality Metrics

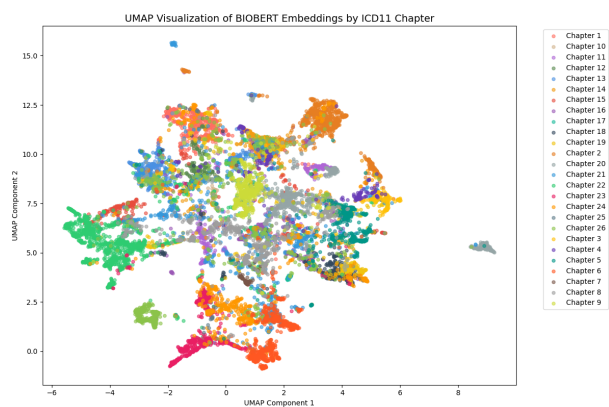
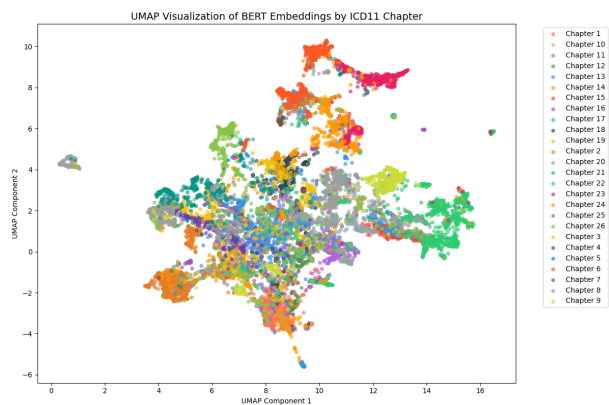
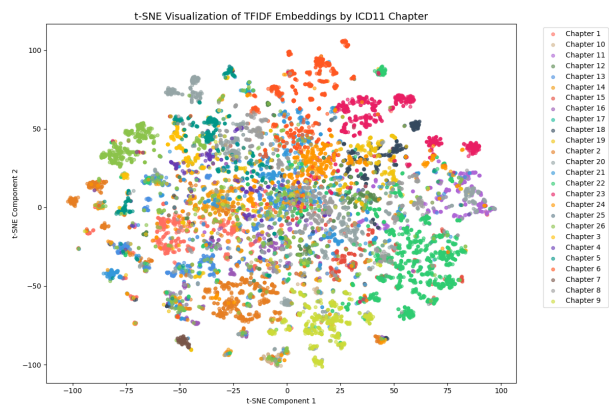
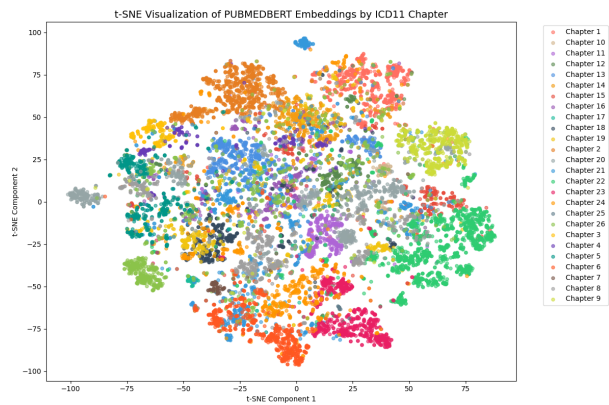
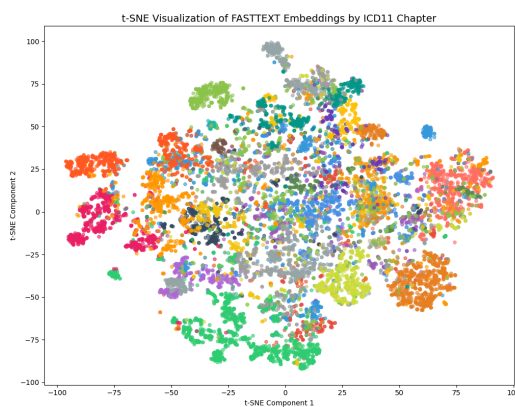
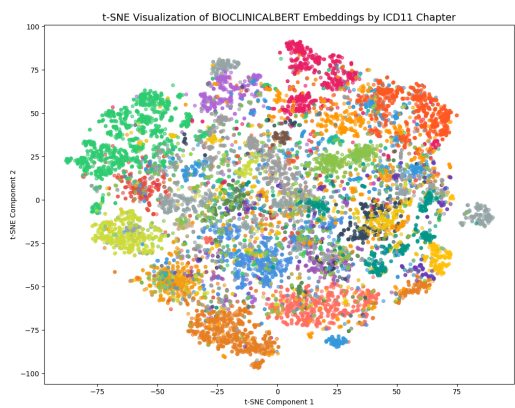
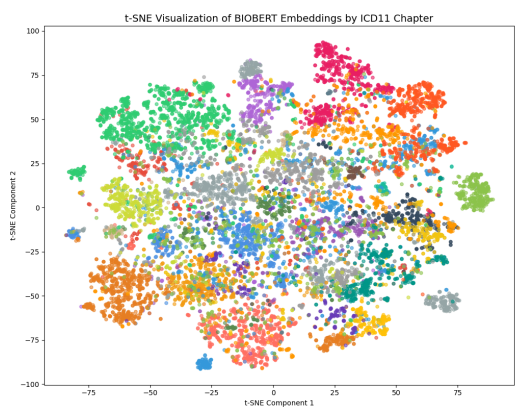
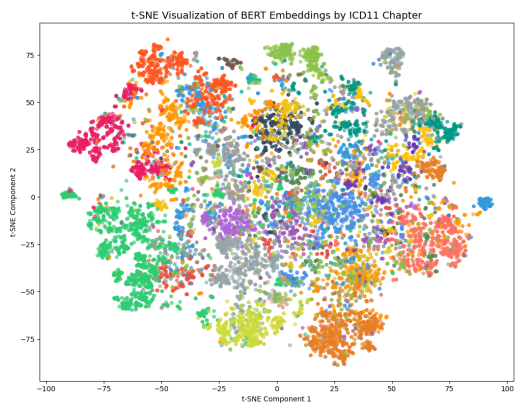
Model	t-SNE Trustworthiness	t-SNE Continuity	UMAP Trustworthiness	UMAP Continuity
BERT	0.9327	0.9001	0.9164	0.9148
BioBERT	0.9128	0.9004	0.8982	0.9093
BioClinicalBERT	0.9180	0.8908	0.8944	0.9041
PubMedBERT	0.9234	0.8974	0.9068	0.9100
TF-IDF	0.7656	0.8148	0.7666	0.8202
FastText	0.9487	0.9273	0.9376	0.9323
GatorTron	0.9224	0.9052	0.9181	0.9193

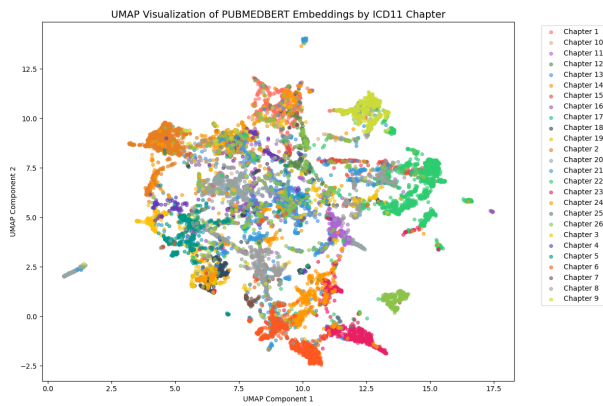
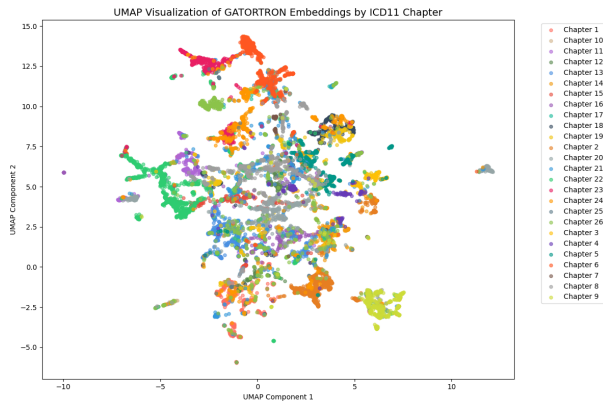
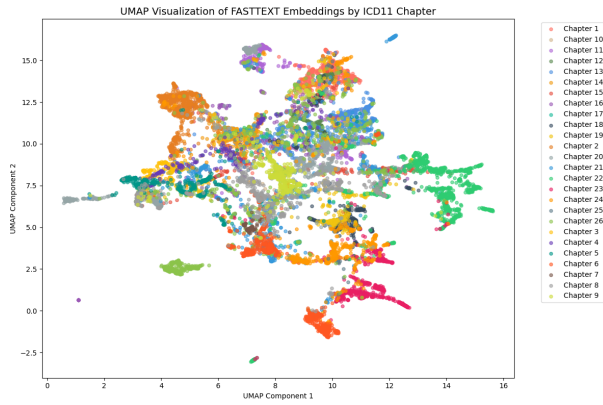
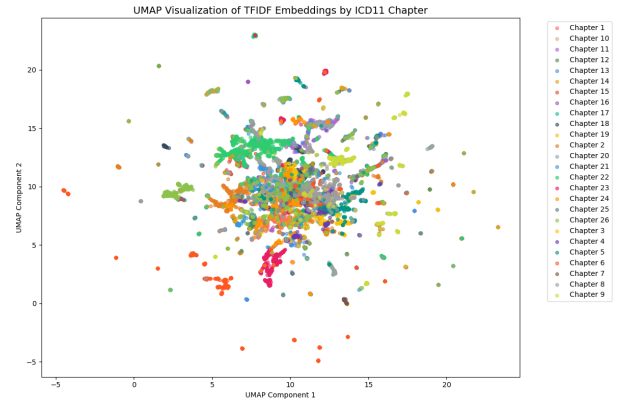
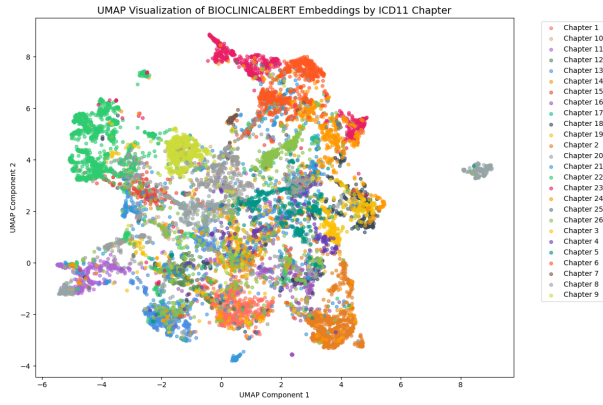
Table 3: Dimensionality Reduction Quality Metrics

Model	t-SNE Silhouette Score	UMAP Silhouette Score
BERT	-0.2187	-0.1551
BioBERT	-0.2033	-0.1698
BioClinicalBERT	-0.1910	-0.1780
PubMedBERT	-0.1315	-0.0836
TF-IDF	-0.3374	-0.2890
FastText	-0.1195	-0.1406
GatorTron	-0.0738	-0.0936

Table 4: Low-Dimensional Clusters Silhouette Scores

C.2 Low-Dimensional Visualizations





C.3 Comorbidity Score Benchmark

Comorbidity Benchmark Score is the Pearson correlation coefficient between the comorbidity scores of disease pairs and the cosine similarity of their embeddings generated by a particular model. See Table 1 for full results.

C.4 Symptom-Disease Matching

See Table 5 for full results.

Model	Mean	Standard Deviation
TF-IDF	0.6158	0.1340
FastText	0.9754	0.0101
BERT	0.9600	0.0137
BioBERT	0.9791	0.0076
BioClinicalBERT	0.9768	0.0068
PubMedBERT	0.9952	0.0017
GatorTron	0.9648	0.0138

Table 5: Mean and standard deviation of cosine similarity of top disease across symptoms

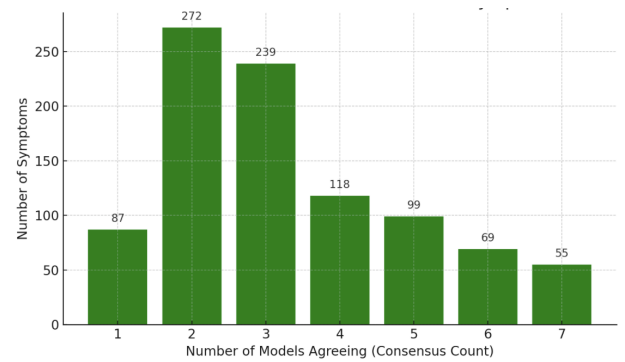


Figure 3: Distribution of consensus count across models

	tfidf	fasttext	bert	biobert	bioclinicalbert	pubmedbert	gatortron
tfidf	100.00	21.73	17.25	17.89	16.08	20.16	22.68
fasttext	21.73	100.00	25.45	27.48	21.09	29.29	26.09
bert	17.25	25.45	100.00	34.61	30.67	31.52	30.67
biobert	17.89	27.48	34.61	100.00	31.31	35.46	33.76
bioclinicalbert	16.08	21.09	30.67	31.31	100.00	26.30	26.94
pubmedbert	20.16	29.29	31.52	35.46	26.30	100.00	38.02
gatortron	22.68	26.09	30.67	33.76	26.94	38.02	100.00

Figure 4: Agreement matrix (% of exact matches)

Useful definitions:

- **Top Disease for a Symptom:** For a given symptom, the top disease is the one whose embedding has the highest cosine similarity to the symptom’s embedding. It is interpreted as the disease most strongly associated with that symptom according to the model.
- **Consensus Count:** The number of symptoms for which exactly n models predicted the same top disease. It reflects how often a specific level of agreement occurs among the models.
- **Agreement Matrix:** A symmetric matrix that shows, for each pair of models, the percentage of symptoms for which both models predicted the same top disease. It measures how similarly different models behave in the symptom–disease matching task.

person. Vary the tone of voice, age group, gender, and social background across the examples. Include formal and informal styles, different levels of education, slang, non-native English speakers, and both brief and detailed descriptions. Do not explain or label the categories in the output. Just return a CSV-formatted list with three columns: prompt, person and category. Each row should contain one prompt and its corresponding category label. There should be 300 rows total (30 per category). Do not include any additional commentary or explanation."

We used each of our 7 models to embed all the generated prompts. Based on cosine similarity, we identified the three most similar ICD-11 codes for each embedding and retrieved the corresponding disease titles. We then checked whether the keyword representing the prompt’s medical category (e.g., "burn," "headache," etc.) appeared in at least one of the top three titles. If it did, we considered the model successfully identifying a relevant disease for that prompt.

C.5 Encyclopedia definition benchmark

Model	4 Symbols	3 Symbols	2 Symbols
TF-IDF	1.06%	8.47%	15.61%
FastText	0.79%	2.12%	4.23%
BERT (base)	51.85%	64.81%	72.49%
BioBERT	66.14%	77.51%	82.01%
BioClinicalBERT	54.50%	65.34%	70.90%
PubMedBERT	75.13%	83.33%	86.24%
GatorTron	2.91%	11.11%	26.19%

Table 6: Prefix match accuracy for 4, 3, and 2 symbol lengths

Model	1 Symbol	No Match	Overall Match
TF-IDF	31.48%	68.52%	31.48%
FastText	16.67%	83.33%	16.67%
BERT (base)	83.07%	16.93%	83.07%
BioBERT	89.95%	10.05%	89.95%
BioClinicalBERT	82.54%	17.46%	82.54%
PubMedBERT	92.33%	7.67%	92.33%
GatorTron	50.79%	49.21%	50.79%

Table 7: Prefix match accuracy for 1 symbol length and overall performance

C.6 Non-medical prompt benchmark

We created this benchmark by picking 10 common illnesses and building 30 different human-like prompts to imitate a scenario in which our models were used to process human prompts and give back the most likely disease code based on the given prompt. This was the ChatGPT prompt used to generate non-medical prompts:

"Generate 30 distinct natural language prompts for each of the following medical categories: 1. burn 2. headache 3. respiratory 4. sprain 5. gastro 6. dermatitis 7. laceration 8. anxiety 9. dorsalgia 10. infection Each prompt should sound like it was written by a different

Model	Top-3 Matches	Acc.	Avg Sim.	Min Sim.	Max Sim.
BioClinicalBERT	21	7.0%	0.885	0.857	0.926
BERT	19	6.3%	0.665	0.580	0.752
BioBERT	14	4.7%	0.891	0.866	0.914
PubMedBERT	6	2.0%	0.970	0.958	0.980
TF-IDF	4	1.3%	0.467	0.360	0.638
FastText	0	0.0%	0.438	0.437	0.438
GatorTron	0	0.0%	0.939	0.885	0.964

Table 8: Model performance based on keyword matching between prompt text and retrieved disease titles